The Light Dark Matter Experiment

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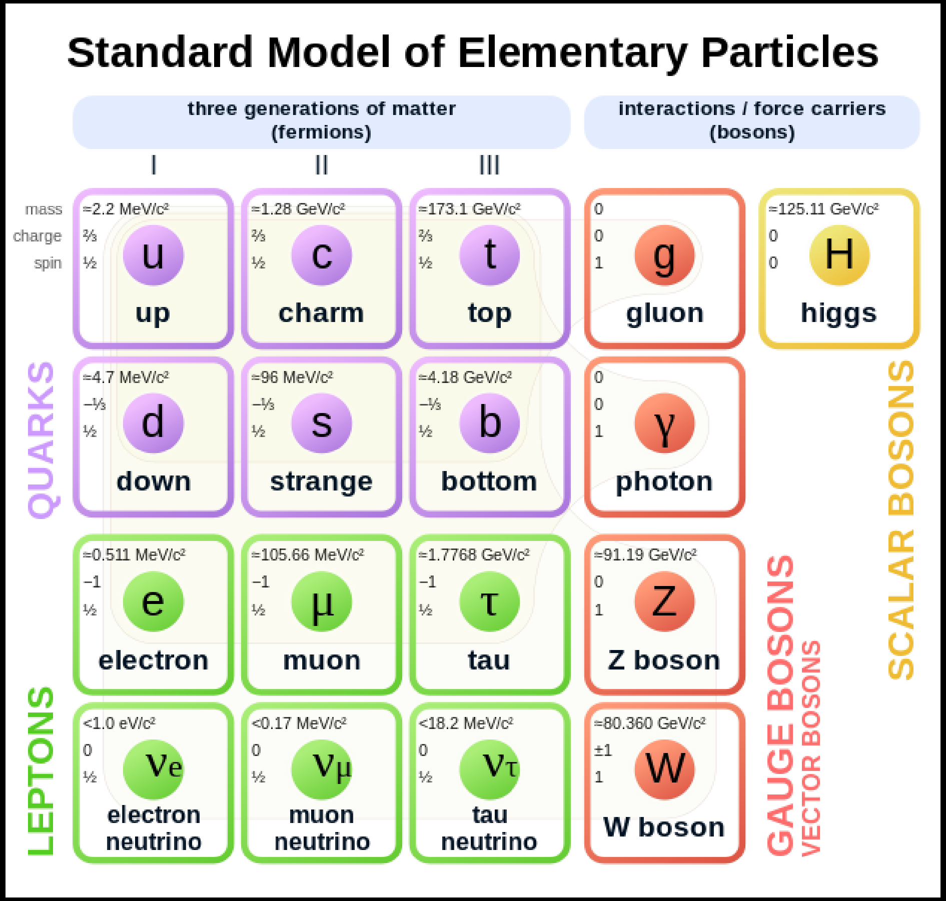
The Standard Model… and a problem



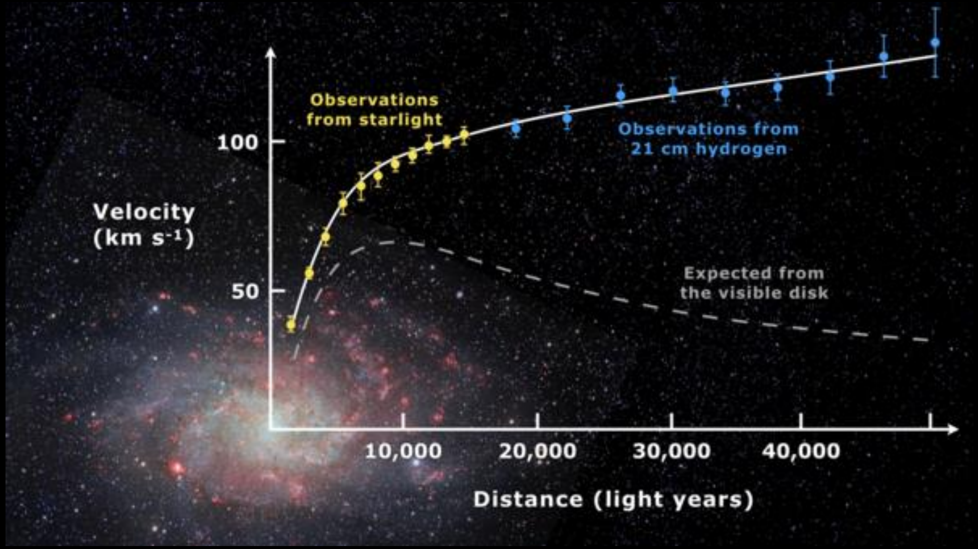
https://en.wikipedia.org/wiki/File:Standard_Model_of_Elementary_Particles.svg

- Describes matter and forces in terms of fundamental particles and their interactions
- Most successful scientific theory *ever developed*!
- Thousands of predictions have been verified by experiment

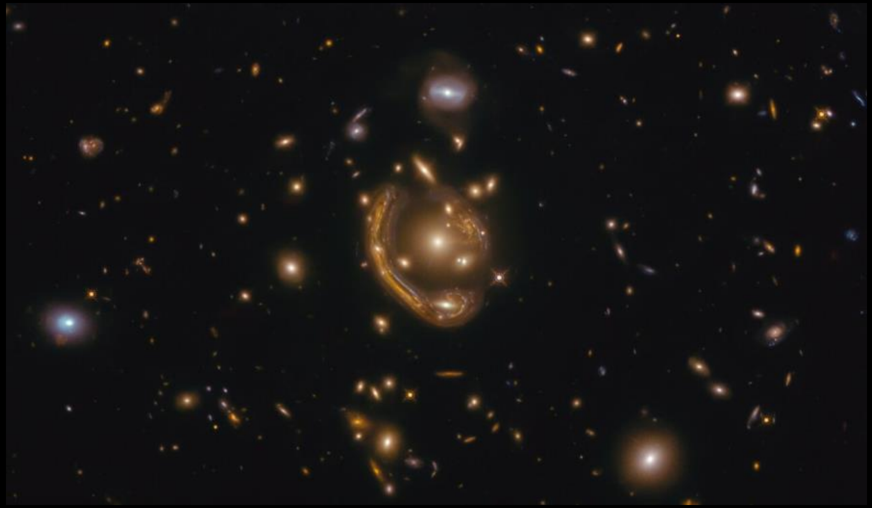
However, several glaring discrepancies between SM and reality!

Dark Matter

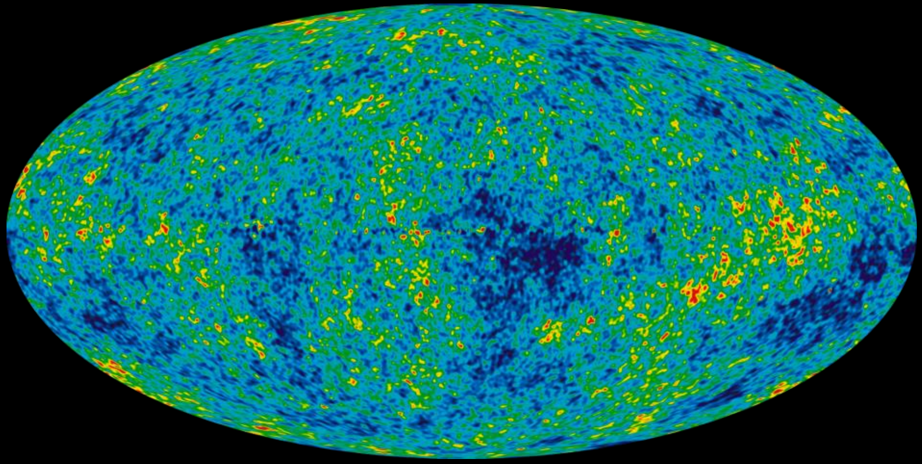
- Strong evidence for *missing mass*, gravitational effects with no visible cause!



Galaxy Rotation Curves

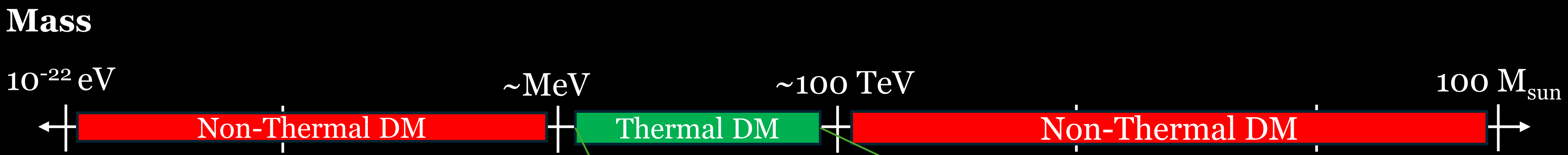


Gravitational Lensing



Cosmic Microwave Background

Thermal Dark Matter



- Candidate masses span <u>~70 orders of magnitude!</u>

- Lots of open parameter space ☺
- The focus of LDMX

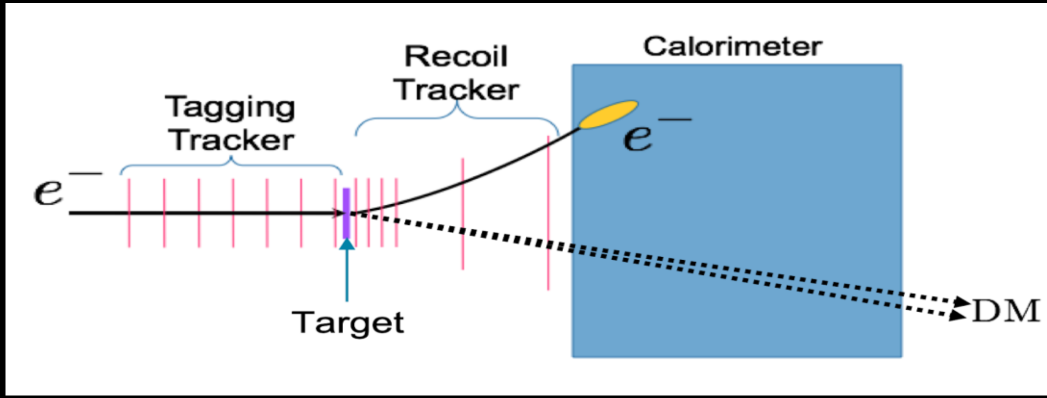
- Restricted parameter space ☹

<u>Simplifying assumption:</u> Dark matter was in thermal equilibrium in the early universe

- Well motivated
- Dramatically reduced parameter space

LDMX Concept -> *Missing Momentum*

- Light DM could theoretically interact with SM matter via a "dark photon" (A')
- Potential interaction: Dark Bremsstrahlung
  - Electron slows down around atom and emits an A' instead of a photon!

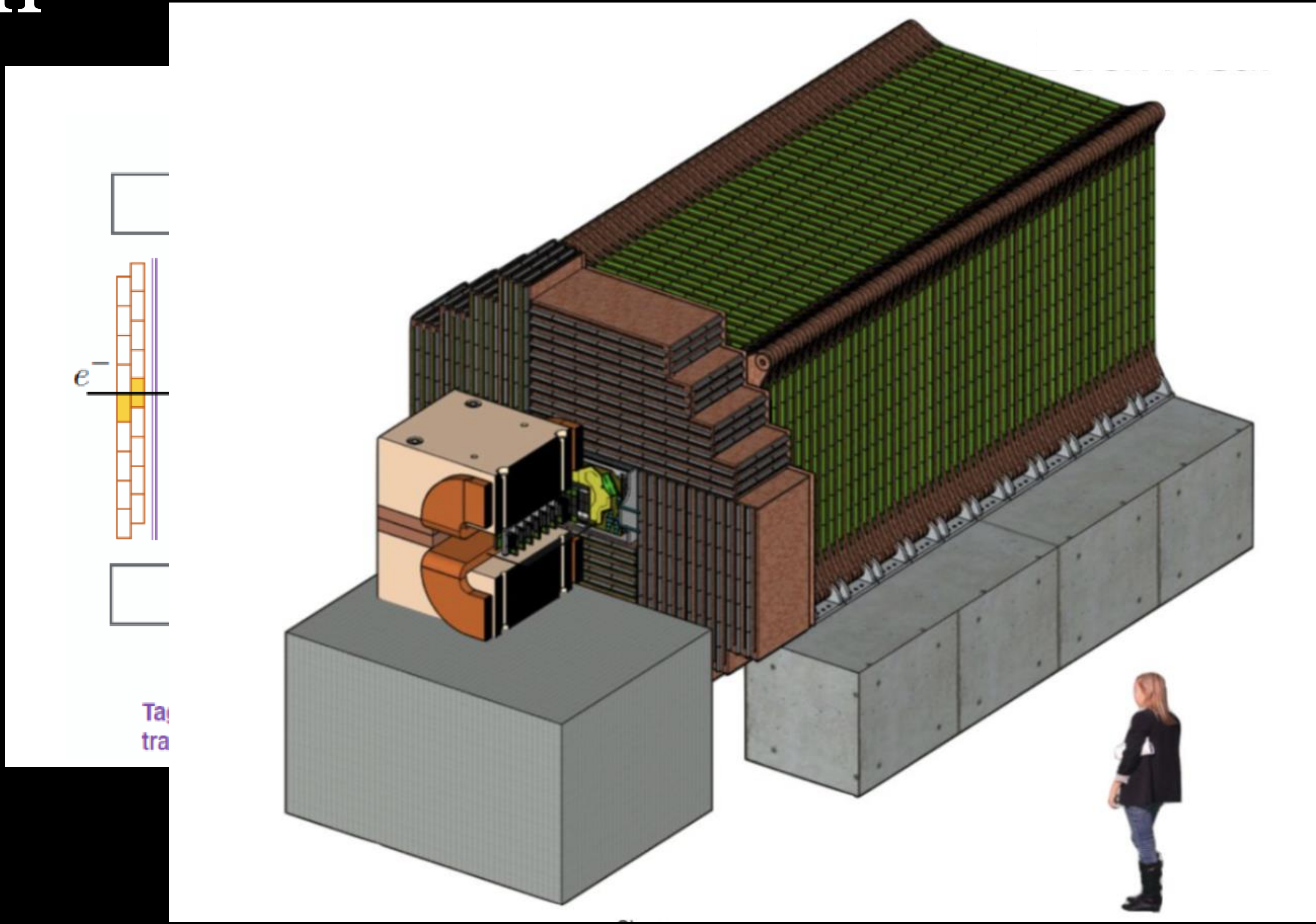


If $m_{A'} > 2m_{DM}$ then A' decays *invisibly*

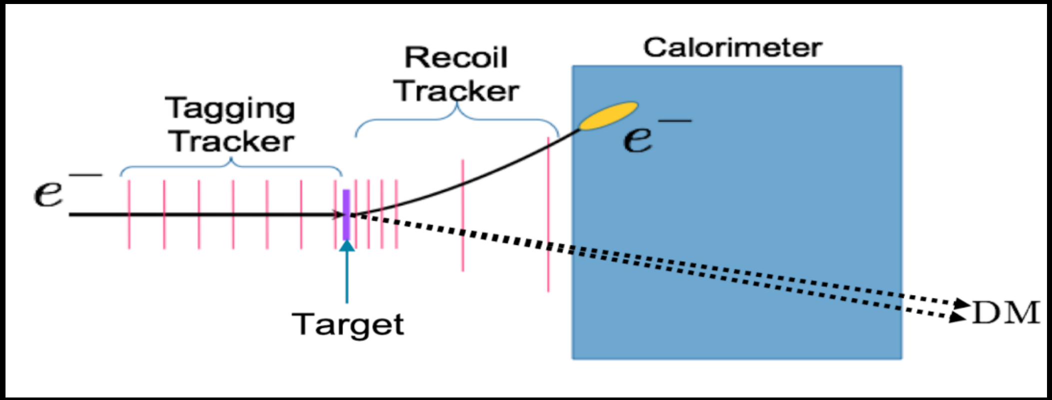
- **Signal Event Signature:**
  - single low energy electron with "missing" energy momentum lost to the A'

LDMX Design

- High-rate electron beam (from SLAC) at Tungsten target to maximize potential dark-brem events
- Series of detectors and calorimeters to carefully look for <u>missing energy and momentum</u> and veto backgrounds

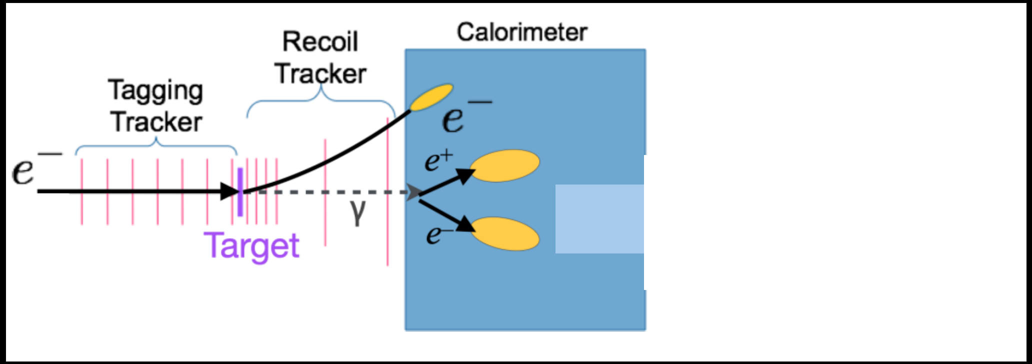


Backgrounds

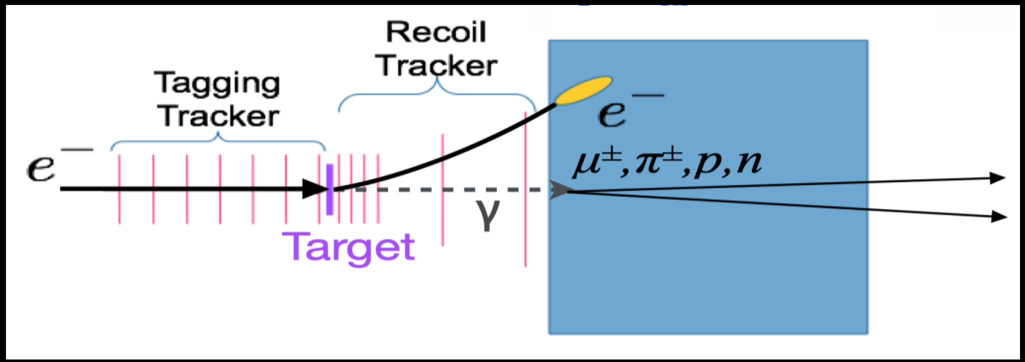


Signal Process

- Signal process can be mimicked by normal brem-events and photonuclear events!
- Ecal, Hcal, and other detectors designed to veto these backgrounds
  - Threshold cuts
  - Boosted decision trees



Brem-Background

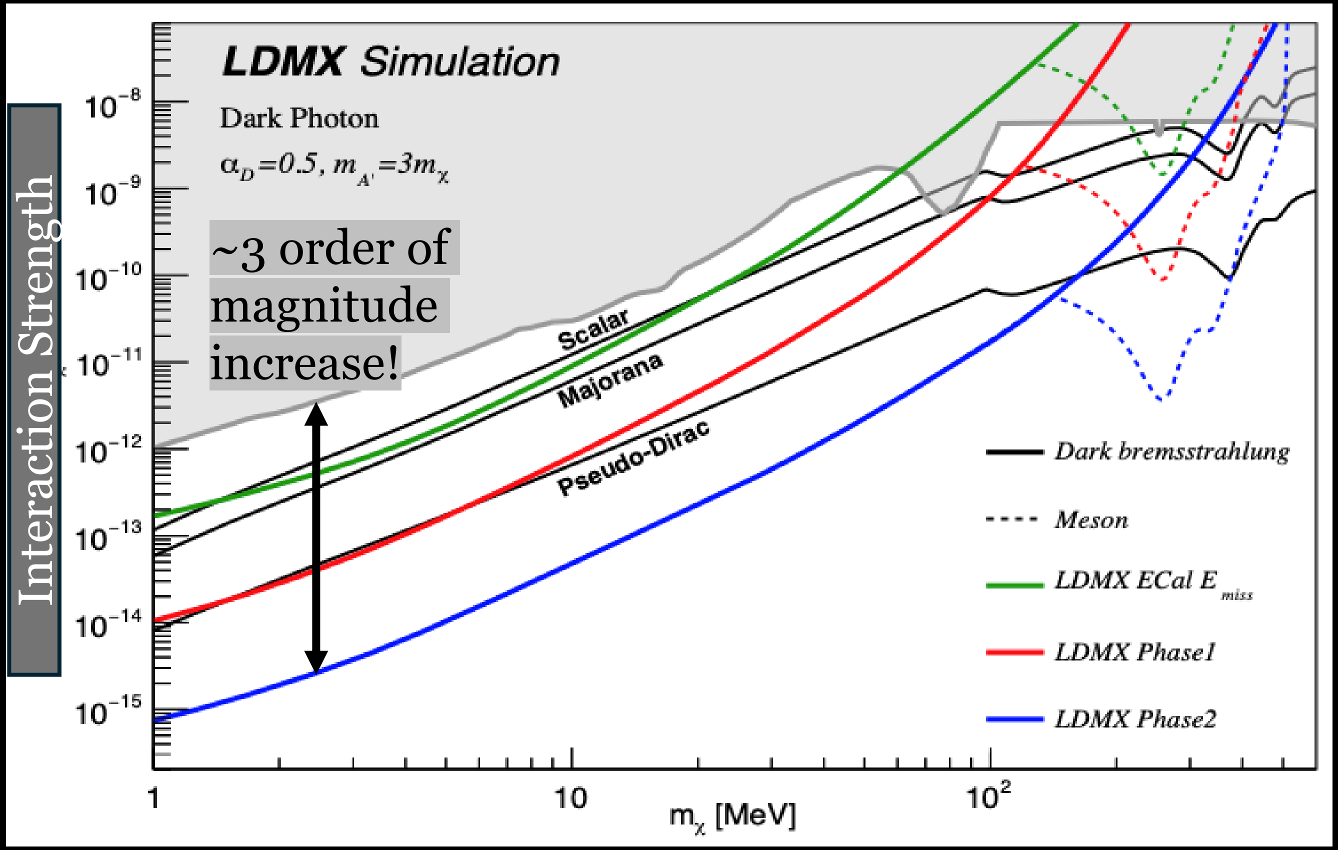


Photonuclear-Background

With all veto systems for ~10^{14} EoT

- <1 background event (!!!)
- ~30-50% signal efficiency

Missing Momentum Sensitivity



https://arxiv.org/abs/2203.08192

And… LDMX has the potential to explore even more physics!

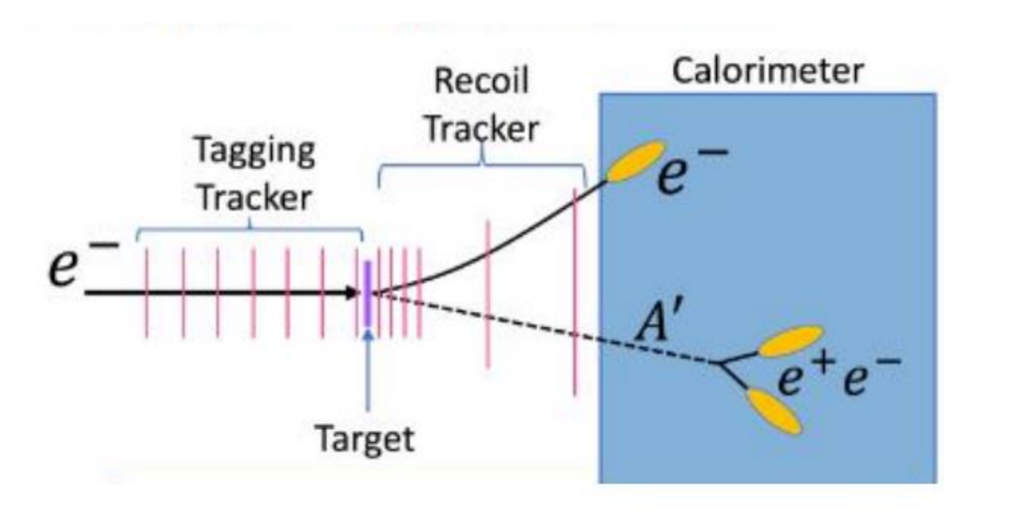
- Within a few weeks of operation, LDMX's missing momentum search will have world-leading results
  - Grey shade is excluded by current experiments
- Due to beam-line delays, most data will be taken at the phase-2 marker in blue.
- Will be huge step forward exploring the Light DM parameter space!

LDMX Visibles -> A Parallel Search

- **What if $m_{A'} < 2m_{DM}$??**
  - A' will decay visibly into SM particles (e^+e^- pair)
  - New signal: sudden appearance of energy deeper in Ecal or Hcal with no sign of other SM particles except soft e^-

	$\text{ECal}_{\text{back half}}$	HCal	$\text{ECal}_{\text{back half}}$ + HCal
ECal	>2.5 GeV	<2.5 GeV	–
HCal	<2.5 GeV	>2.5 GeV	–
Intermediate	<2.5 GeV	<2.5 GeV	>2.5 GeV

My research focus!



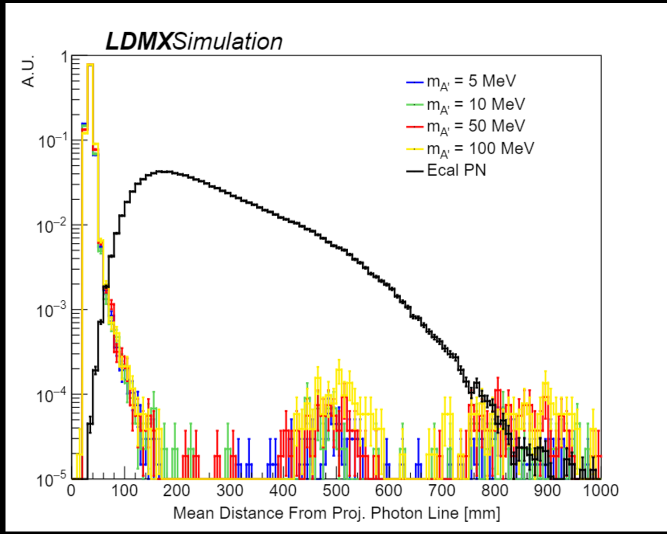
- Background processes become more difficult to veto in visible decays
  - Rely on Boosted Decision Trees (BDTs) in three distinct energy ranges to parse signal and background

Intermediate Region BDT

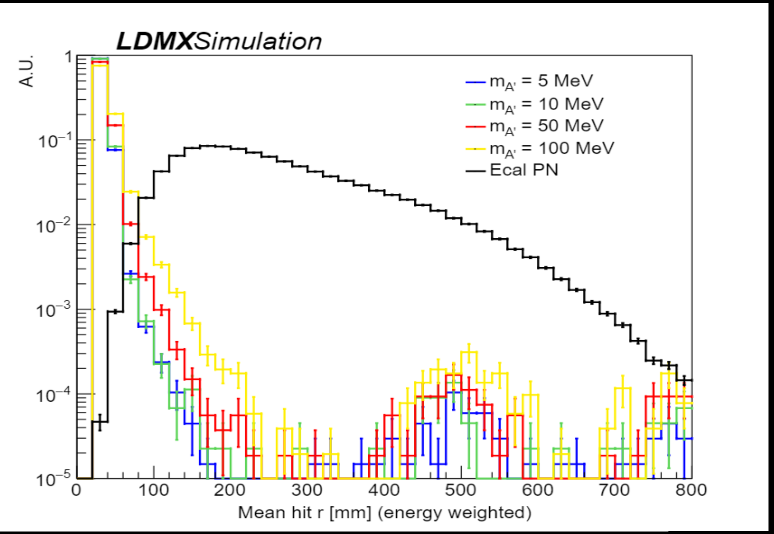
- **Discriminating Features**
  - Some observables have clear difference between signal and background -> Good!
  - Others signal and background very similar -> Bad!

- Could make threshold cuts on single features with good discrimination…

Or even better -> combine multiple high performing features with BDT!



Good Input Feature ☺

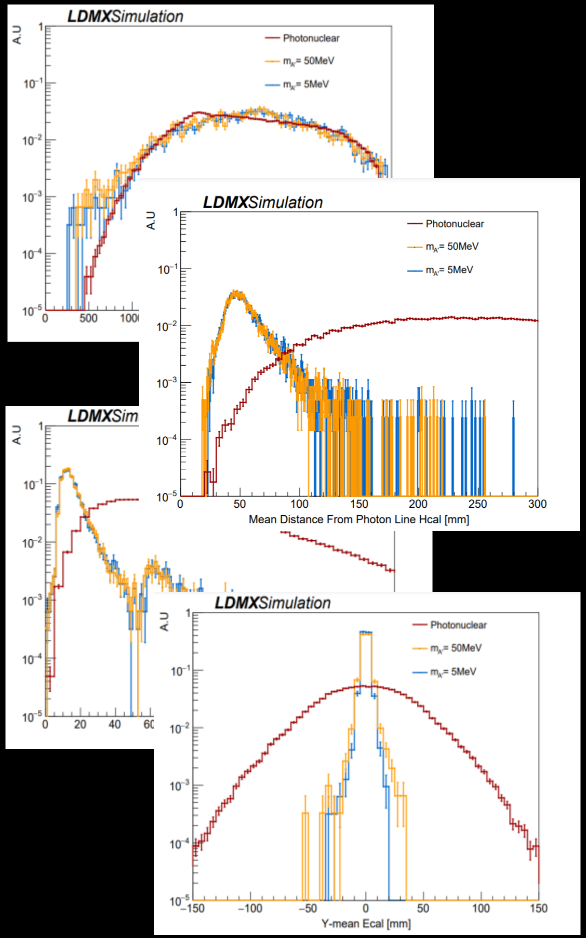


Bad Input Feature ☹

Intermediate Region BDT

One remaining background to consider, see Anmol's talk next!

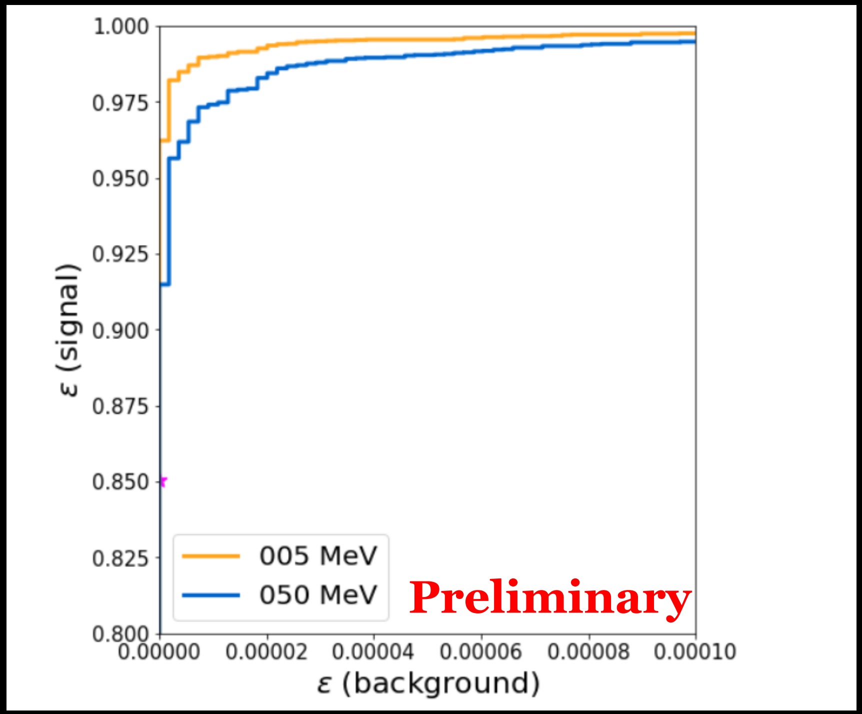
- **BDT combines all our input features for highly effective signal identification**



19 input features

(only a few shown)

BDT



Output ROC Curve

Promising *preliminary* results:

- Only 5 background events for ~10^{14} EoT on 20% signal efficiency

Conclusion

- LDMX Missing Momentum search will improve Light DM sensitivity by several orders of magnitude
- LDMX Visibles offers complementary search channel with potential for low background and competitive signal efficiency

Thank You!

Questions?

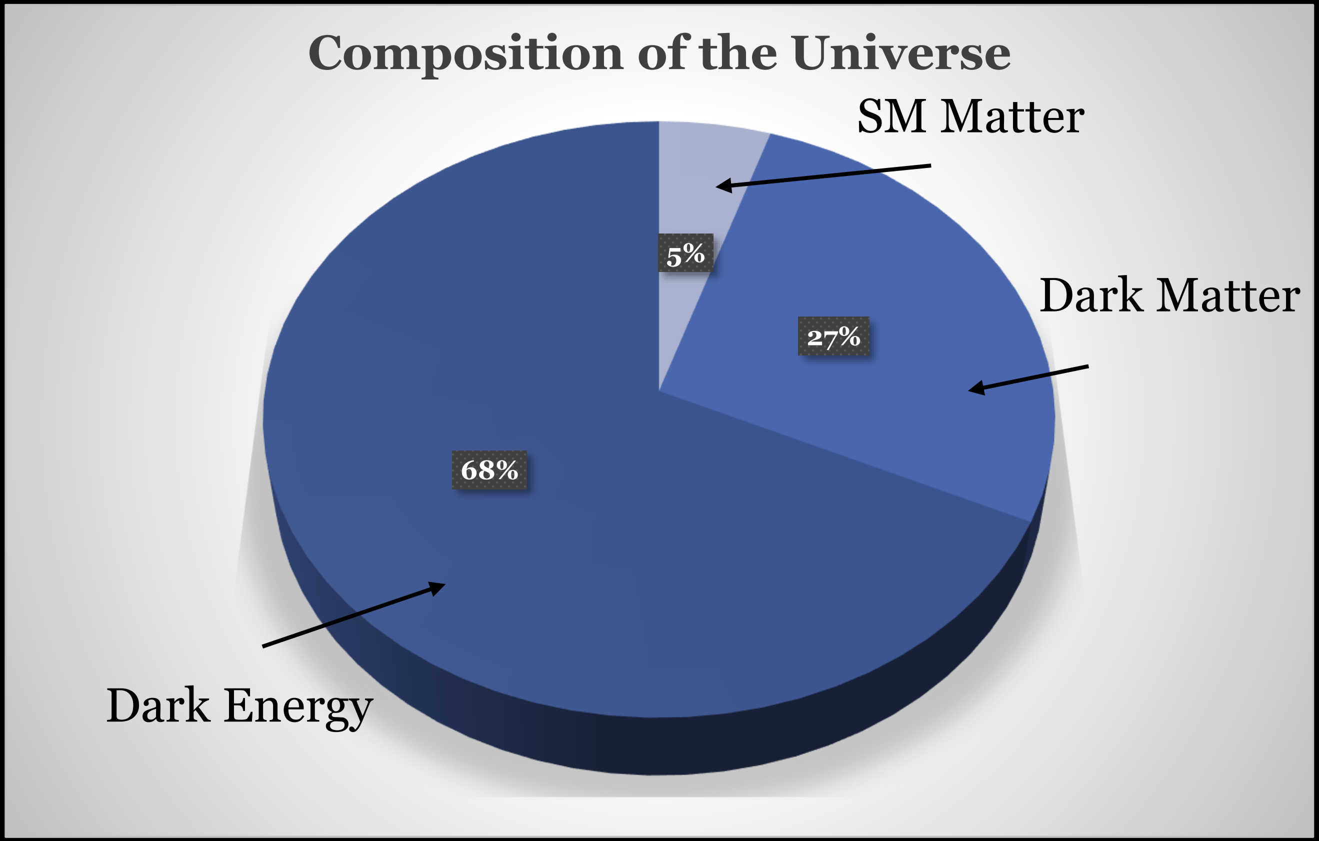
State-of-the-art LDMX prototype

Caltech
Fermilab
Lunds Universitet
University of Minnesota
UCSB University of California Santa Barbara
Carnegie Mellon University
SLAC National Accelerator Laboratory
Stanford University
Texas Tech University
University of Virginia

Our International Collaboration

Dark Matter

- Standard Model only describes only 5% of the energy content of the universe
- Discovering the identity of the dark matter would be a huge step forward in understanding of the universe!



Dark energy is a story for a different day!

Image Sources

- https://en.wikipedia.org/wiki/File:Standard_Model_of_Elementary_Particles.svg
- https://en.wikipedia.org/wiki/Galaxy_rotation_curve#/media/File:Rotation_curve_of_spiral_galaxy_Messier_33_(Triangulum).png
- https://esahubble.org/wordbank/gravitational-lensing/
- https://en.wikipedia.org/wiki/Cosmic_microwave_background#/media/File:Ilc_9yr_moll4096.png
- https://1000logos.net/virginia-cavaliers-logo/